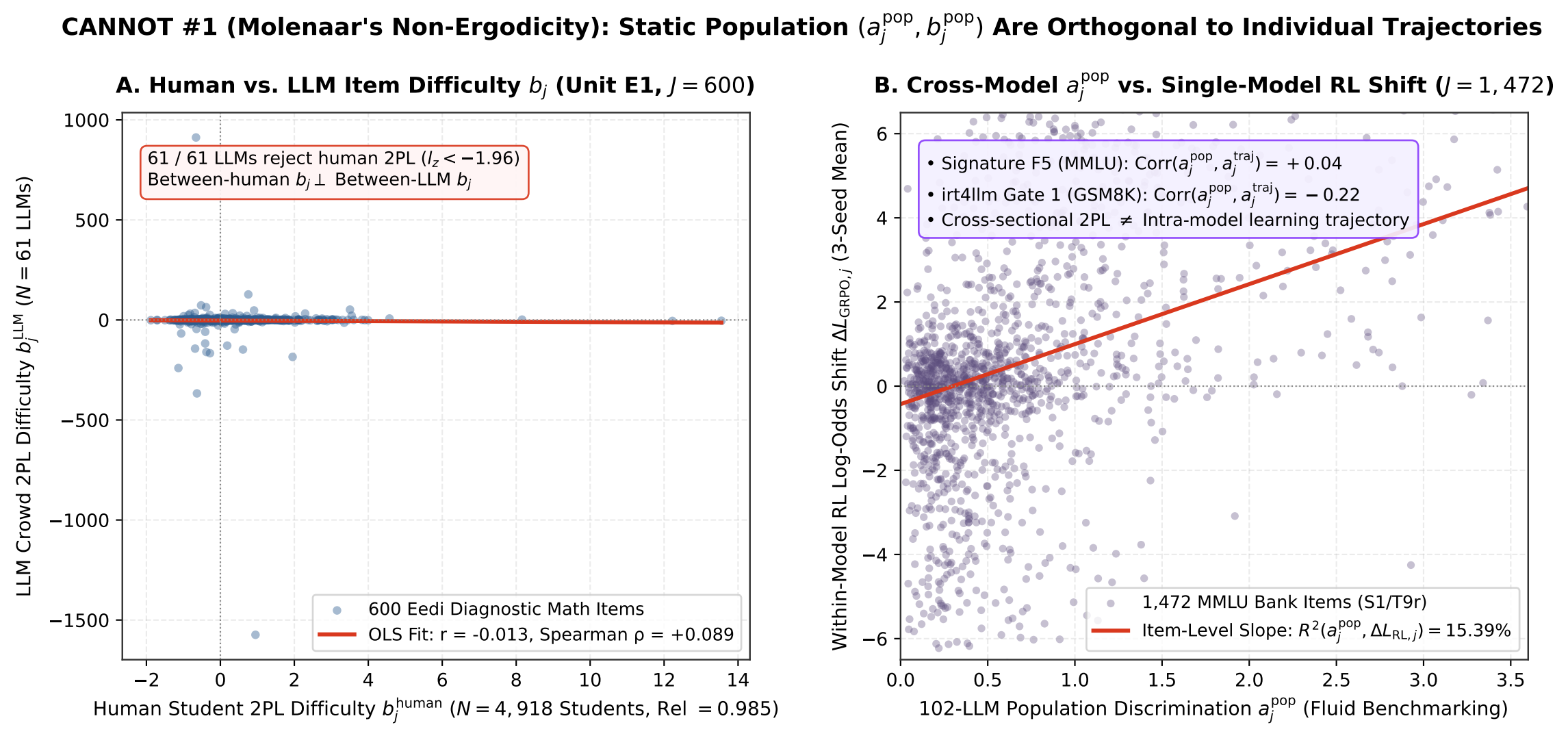

CANNOT #1 (Molenaar's Non-Ergodicity): Static Population (a_j^{pop}, b_j^{pop}) Are Orthogonal to Individual Trajectories
A. Human vs. LLM Item Difficulty b_j (Unit E1, $J = 600$)
61 / 61 LLMs reject human 2PL ($l_z < -1.96$)
Between-human $b_j \perp$ Between-LLM b_j
600 Eedi Diagnostic Math Items
OLS Fit: r = -0.013, Spearman ρ = +0.089
LLM Crowd 2PL Difficulty b_j^{LLM} ($N = 61$ LLMs)
Human Student 2PL Difficulty b_j^{human} ($N = 4,918$ Students, Rel $= 0.985$)
B. Cross-Model a_j^{pop} vs. Single-Model RL Shift ($J = 1,472$)
• Signature F5 (MMLU): Corr$(a_j^{pop}, a_j^{traj}) = +0.04$
• irt4llm Gate 1 (GSM8K): Corr$(a_j^{pop}, a_j^{traj}) = -0.22$
• Cross-sectional 2PL $\neq$ Intra-model learning trajectory
1,472 MMLU Bank Items (S1/T9r)
Item-Level Slope: $R^2(a_j^{pop}, \Delta L_{RL,j}) = 15.39\%$
Within-Model RL Log-Odds Shift $\Delta L_{GRPO,j}$ (3-Seed Mean)
102-LLM Population Discrimination a_j^{pop} (Fluid Benchmarking)